VIRTUAL CLOUD COMPUTING | FINAL PROJECT

RL-Based Auto-Scaling for Hospital Cloud

Reducing SLA violations by 95.2% using Proximal Policy Optimization on AWS

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95.2%

SLA Violation Reduction

18.1%

Cost Reduction

THE PROBLEM

OPD Rush

9am-12pm: CPU hits 80-90%. AWS reacts AFTER the spike — patients experience crashes.

Idle Waste

3am-6am: CPU at 10-20%. VMs keep running unnecessarily — money wasted every night.

Spin-up Delay

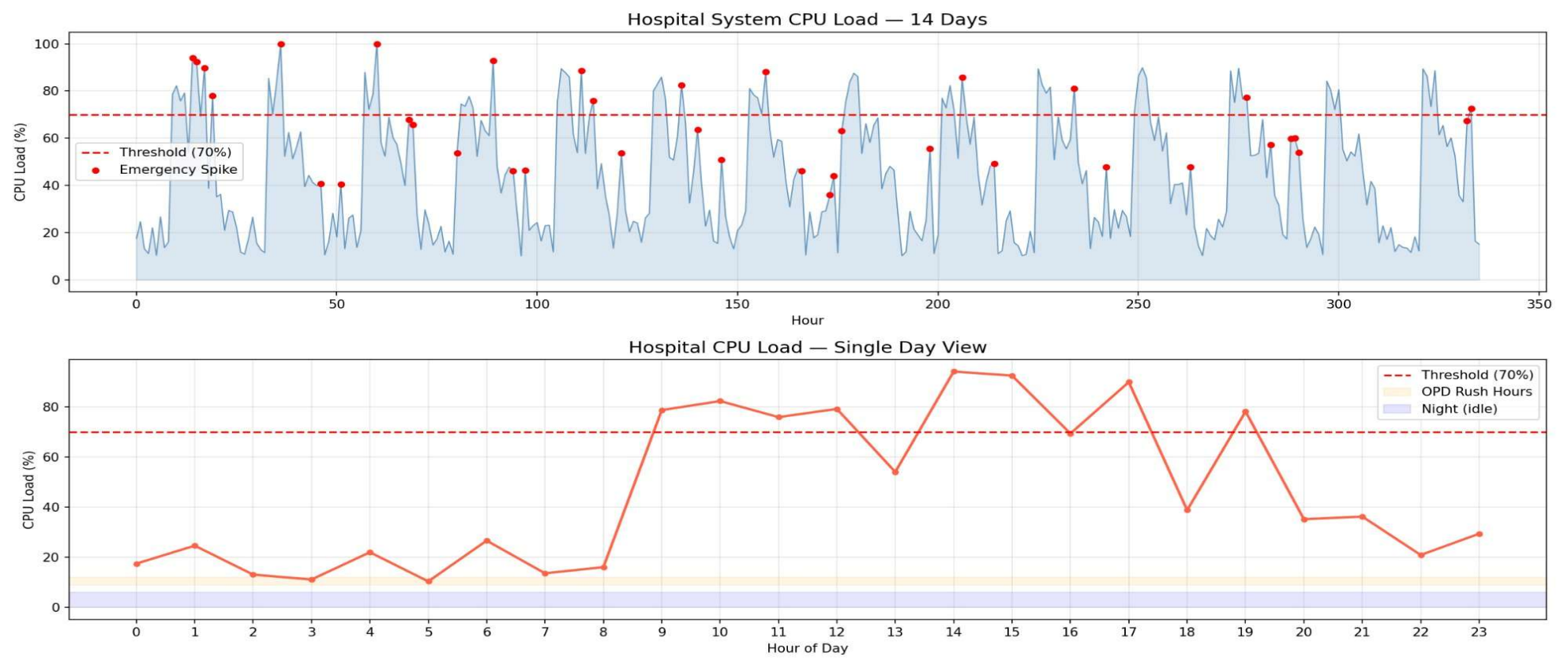
New VM takes 3+ hours to activate. SLA violated during this window every single day.

No Learning

AWS threshold rule never learns. Makes the same mistakes on Monday as it did last Monday.

In healthcare, SLA violations = delayed treatment = risk to life

HOSPITAL WORKLOAD — 14 DAYS OF REAL PATTERNS



336 hours simulated • 38 emergency spikes • 69 hours above SLA threshold • Peak CPU: 100%

OUR SOLUTION

Replace AWS threshold rule with a PPO Reinforcement Learning agent that LEARNS and PREDICTS

AWS Default

- ✗ Waits for CPU > 70%
- ✗ Reacts AFTER damage done
- ✗ 3-hour spin-up delay
- ✗ No memory of patterns
- ✗ Same mistakes daily

→

RL Agent (PPO)

- ✓ Predicts BEFORE spike hits
- ✓ Proactive — scales early
- ✓ Pre-warms VMs in advance
- ✓ Learns OPD rush patterns
- ✓ Gets smarter every episode

VCC CONCEPTS DEMONSTRATED

Virtualization

EC2 VMs created & destroyed dynamically on AWS

Auto-Scaling

RL agent controls horizontal VM scaling

IaaS Model

Raw VM infrastructure managed on AWS EC2

Containerization

Hospital services isolated in Docker containers

Cost Optimization

18.1% cost reduction through intelligent scaling

Elasticity

System grows during OPD rush, shrinks at night

SLA Management

Response time enforced under 70% CPU threshold

Cloud Monitoring

AWS CloudWatch feeds live metrics to RL agent

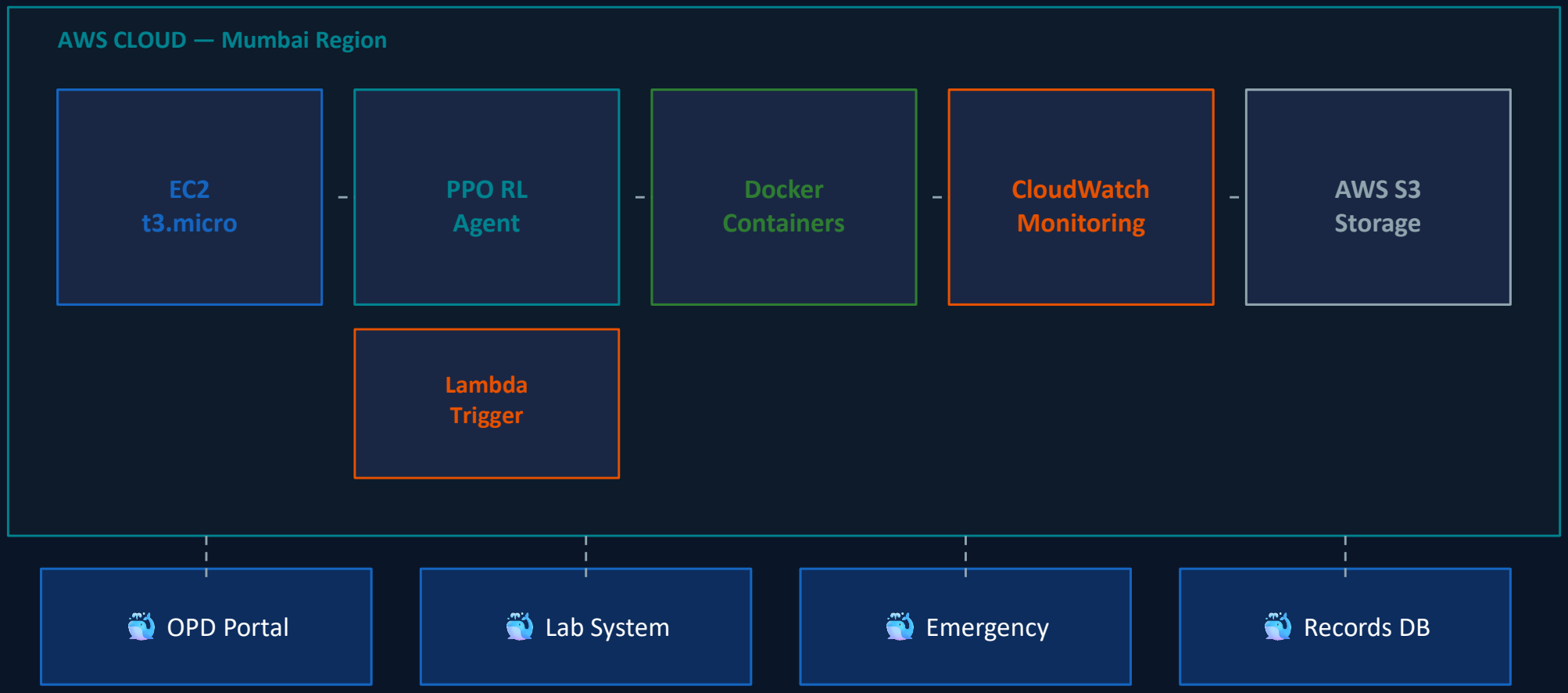
Serverless / FaaS

AWS Lambda triggers scaling decisions

Multi-tenancy

Each hospital department = isolated Docker tenant

SYSTEM ARCHITECTURE ON AWS



BASELINE — AWS THRESHOLD SCALER (The Problem)

42

SLA Violations

666

VM Hours Used

$6.93

14-Day Cost

1.98

Avg VMs

Reactive scaling causes 42 SLA violations in just 14 days — predictable, preventable, unacceptable

RL AGENT TRAINING — 50,000 Steps of Learning

10% Random — 69 violations

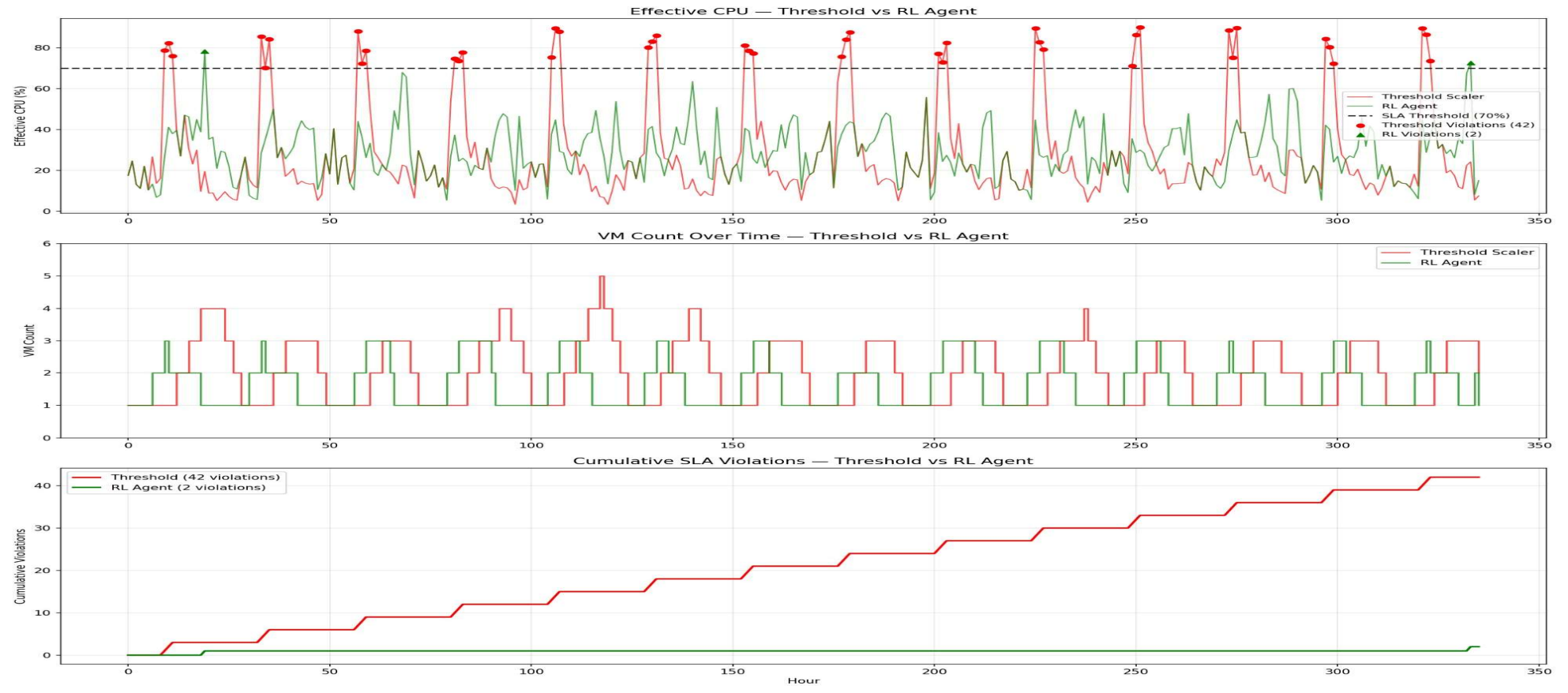
30% Learning patterns

50% Converging fast

100% Only 2 violations!

Reward rises from -25 to +249 as agent masters hospital workload patterns

RESULTS — RL Agent vs Threshold Scaler



RL Agent: 2 violations vs Threshold: 42 violations — Cumulative SLA curve (bottom) tells the complete story

QUANTITATIVE RESULTS

Metric	Threshold (AWS)	RL Agent (PPO)	Improvement
SLA Violations (14 days)	42	2	95.2% ↓
Total VM Hours	666	546	18.0% ↓
Avg VMs Running	1.98	1.62	18.2% ↓
Estimated Cost (14 days)	$6.93	$5.68	18.1% ↓

RL Agent outperforms AWS default threshold scaling on ALL 4 metrics simultaneously

FUTURE SCOPE

Multi-Cloud Deployment

Extend RL agent to span AWS + Azure + GCP for cross-cloud cost optimization

Kubernetes Orchestration

Replace Docker Compose with K8s for production-grade container management

Federated RL

Multiple hospitals collaboratively train a shared RL policy without sharing data

Real MIMIC-III Data

Validate results on authentic ICU patient admission timestamps from PhysioNet

Explainable RL

Apply SHAP to understand which state features drive the agent's scaling decisions

Predictive CloudWatch

Use AWS Anomaly Detection to feed predictive signals into the RL state space

CONCLUSION

A PPO-based Reinforcement Learning agent reduces SLA violations by 95.2% and cloud infrastructure cost by 18.1% compared to AWS default threshold-based auto-scaling — demonstrating that intelligent, proactive cloud resource management is both technically feasible and clinically significant for hospital systems.

95.2%

SLA Violations Reduced

18.1%

Cloud Cost Saved

10

VCC Concepts Proven